Ever. It's as if the measurement itself forces the electron to gather together all its spinning motion and direct it to be either clockwise or counterclockwise about the axis you happened to have focused on. Moreover, because of your measurement's influence on the electron's spin, you lose the ability to determine how it was spinning about a horizontal axis, about a back-and-forth axis, or about any other axis, prior to your measurement. These features of quantum mechanical spin are hard to picture fully, and the difficulty highlights the limits of classical images in revealing the true nature of the quantum world. Nevertheless, the mathematics of quantum theory, and decades of experiment, assure us that these characteristics of quantum spin are beyond doubt.

The reason for introducing spin here is not to delve into the intricacies of particle physics. Rather, the example of particle spin will, in just a moment, provide a simple laboratory for extracting wonderfully unexpected answers to the reality question. That is, does a particle simultaneously *have* a definite amount of spin about each and every axis, although we can never know it for more than one axis at a time because of quantum uncertainty? Or does the uncertainty principle tell us something else? Does it tell us, contrary to any classical notion of reality, that a particle simply does not and cannot possess such features simultaneously? Does it tell us that a particle resides in a state of quantum limbo, having no definite spin about any given axis, until someone or something measures it, causing it to snap to attention and attain—with a probability determined by quantum theory—one particular spin value or another (clockwise or counterclockwise) about the selected axis? By studying this question, essentially the same one we asked in the case of particle positions and velocities, we can use spin to probe the nature of quantum reality (and to extract answers that greatly transcend the specific example of spin). Let's see this.

As explicitly shown by the physicist David Bohm,[11] the reasoning of Einstein, Podolsky, and Rosen can easily be extended to the question of whether particles have definite spins about any and all chosen axes. Here's how it goes. Set up two detectors capable of measuring the spin of an incoming electron, one on the left side of the laboratory and the other on the right side. Arrange for two electrons to emanate back-to-back from a source midway between the two detectors, such that their spins—